# 1 Convolution and Distribution Functions

## 1.1 Convolution

**Definition 1.1.1 ▸ convolution**

Given two Lebesgue measurable functions $f$ and $g$ on $\mathbb{R}^n$, we define the ***convolution*** $f * g$ of $f$ and $g$ to be the function defined for each $x \in \mathbb{R}^n$ by

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\, dy.$$

**Lemma 1.1.2**

If $f$ is a Lebesgue measurable function on $\mathbb{R}^n$, then the function $F$ defined on $\mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n}$ by

$$F(x, y) = f(x - y)$$

is $\lambda_{2n}$-measurable.

**Theorem 1.1.3**

If $f \in L^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$, and $g \in L^1(\mathbb{R}^n)$, then $f * g \in L^p(\mathbb{R}^n)$ and

$$\|f * g\|_p \leq \|f\|_p \|g\|_1.$$